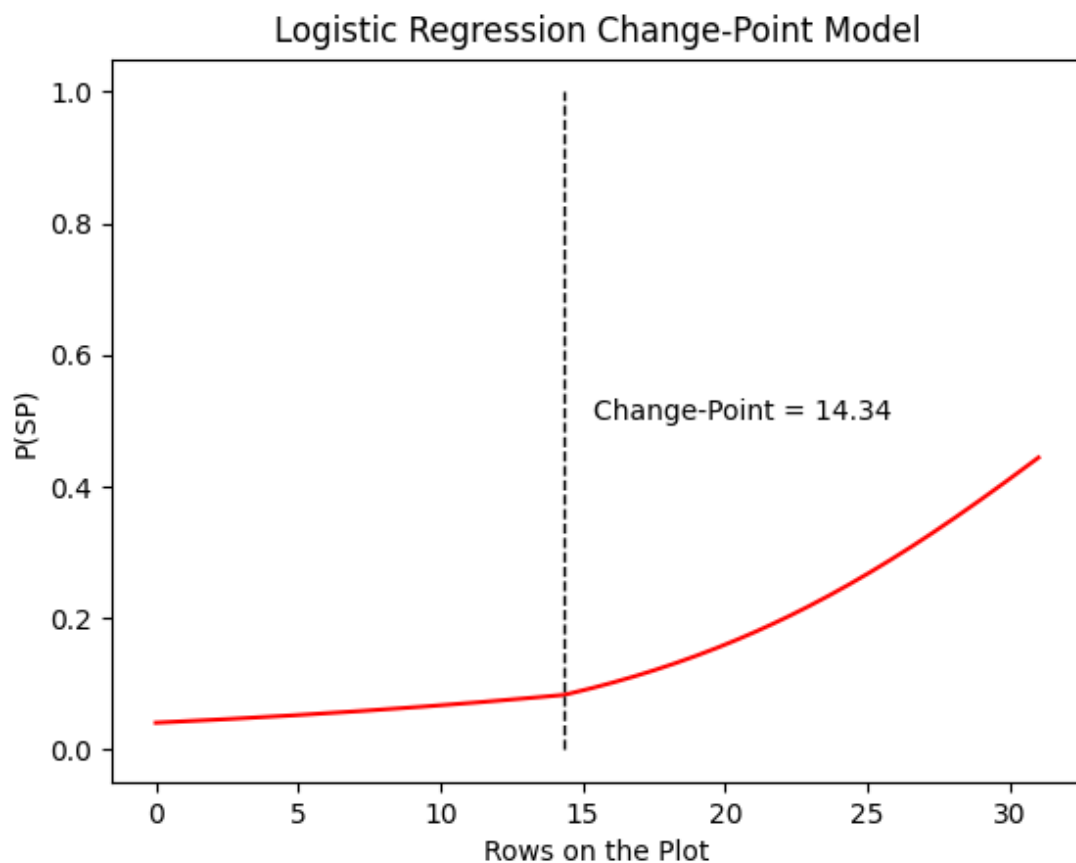


Project Proposal

Introduction

During my time on the tree census team, I have noticed a veritable difference in the distribution of shortleaf pines (SP) versus loblolly pines (LP) in the north and south halves of the plot. In the south half of the plot there is a significantly larger portion of LP. In the north half, SP are more competitive. To quantify where this difference starts, I conducted change-point detection using logistic regression¹. This method can be used to show how biological processes change over space and time². The model showed that at around row 14, the probability of finding SP vs. LP started getting significantly higher.



Hypothesis

This observation lead me to question how competitive SP is in average and total biomass in the separate halves of the plot. I hypothesize that the SP are more competitive in biomass with LP in the north than in the south end of the plot. My assumption is that the higher number of SP gives them more opportunity to get to a bigger size. This research also begs the question of why SP have a higher distribution in the north end of the plot.

Methods

In order to test this hypothesis, I will use allometric equations that take variables such as tree DBH and height to calculate an estimate of trees biomass. To calculate height, I will use arial and terrestrial LiDAR data. I will then find the average biomass of each species and compare the results. Along with this, I plan to create heatmaps of biomass for each species.

Shortleaf Pine Equations

For SP, I have selected 2 equations. Equation one³ requires dbh and the knowledge of if the stand has been previously thinned(x). It's R² value is 0.988. This equation is limited to trees of dbh 7-40 cm. Because of this limiting factor, it will only be used until tree heights can be calculated. Equation two⁴ requires dbh and height of the tree. Height will be calculated using lidar data. Its R² is 0.99. This equation uses pounds and must be converted to metric. The third Equation⁵ I found cannot be used, as it calculates green weight.

Equation One:

$$y = e^{(-1.36184-0.06517x)}(DBH)^{2.134146} + e^{-8.37006}(DBH)^{(3.553264-0.11295x)} + e^{-5.35995}(DBH)^{2.153901-0.10494x}$$

Equation Two:

$$\text{Log}(y) = -1.52244 + 1.11886 \text{Log}(DBH^2 * H)$$

Loblolly Pine Equations

For LP, I have selected 3 equations⁶. Equation one requires dbh. It's R² value is 0.969. Equation 2 requires dbh and height. It's R² is 0.982. Equation 3 requires dbh, stand age, and height. It's R² is 0.985. A separte equation⁷ is more accurate when dbh is under 10cm, but didn't get selected because you must know the age of each tree. Another equation⁸ is also more accurate when dbh is under 10cm, but didn't get selected because you must know if the tree is "dominant" or "suppressed".

Equation One:

$$y = 0.037403(DBH^{2.676835})$$

Equation Two:

$$y = 0.026256(DBH^{2.015144})(H^{0.864052})$$

Equation Three:

$$y = 0.020594(DBH^{2.082279})(H^{1.081437})(AGE^{-0.205654})$$

Relevance

This research will give CBC and Dr. Hogan's Lab access to information pertaining to the nature of the pines on the plot. It will give a numerical and visual measure of the biomass of the pines, and show whether SP have been able to establish themselves in a meaningful way. Showing that SP have established themselves in a meaningful way is important to CBC, as it shows that their management practices have been successful in keeping an element of a healthy piney woods ecosystem.

Materials Needed

For this research, I need access to the plot data, and both terrestrial and arial LiDAR data to calculate biomass. I also need access to the logging history of the plot to aquire the age of the forest stand. The analysis will take place on my personal computer, but depending on the intensity of the computations, I may need access to high-performance computing.

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